

TECHNICAL REPORT

Complete Business Analysis: Customer Churn, Property Pricing & Sales Performance

1. Executive Summary

This project presents an end-to-end business analytics study covering three business areas: **customer churn, property pricing, and sales performance**. The objective was to analyze business data, identify meaningful patterns, apply statistical and machine learning techniques, and translate the findings into practical business recommendations.

Three datasets were analyzed independently: a customer churn dataset containing 500 records, a property dataset containing 300 records, and a sales dataset containing 100 records. The datasets represent different business modules and were therefore not merged row-by-row.

The project used descriptive statistics, exploratory data analysis, correlation analysis, hypothesis testing, logistic regression, and linear regression.

For customer churn, the logistic regression model achieved **93.0% accuracy, 75.0% precision, 54.5% recall, 63.2% F1-score, and 0.982 ROC-AUC** on the test dataset.

For property pricing, the linear regression model achieved an **R^2 of 0.941**, with an MAE of **2,188,736.34** and RMSE of **2,907,633.21**.

For sales performance, total sales were **12,365,048**, with 478 units sold and an average transaction value of **123,650.48**. Laptop recorded the highest total sales among the products, while North recorded the highest total regional sales.

A one-way ANOVA comparing regional sales produced an F-statistic of **2.164** and a p-value of **0.097**. Therefore, there was insufficient statistical evidence at the 5% significance level to conclude that average sales differed across the four regions.

The project concludes with recommendations for customer retention, property pricing, sales optimization, and continuous performance monitoring.

2. Business Problem

Organizations generate large amounts of customer, sales, and operational data. However, raw data alone does not provide useful business value unless it is properly cleaned, analyzed, interpreted, and converted into actionable decisions.

This project addresses three related business analytics problems:

Customer Churn Problem

Customer loss can reduce revenue and increase the cost of acquiring replacement customers. Therefore, businesses need to identify customer characteristics associated with churn and develop methods for identifying customers who may be at higher risk of leaving.

Property Pricing Problem

Property prices can vary according to characteristics such as area, number of bedrooms, bathrooms, property age, location, and property type. A data-driven pricing model can help provide a quantitative reference for evaluating properties.

Sales Performance Problem

Businesses need to understand which products generate higher sales, how sales vary across regions, and how transaction characteristics such as quantity and price relate to total sales.

3. Project Objectives

The main objectives of this project were:

1. Analyze customer data to understand churn-related patterns.
2. Develop a machine learning model to predict customer churn.
3. Analyze property characteristics associated with property prices.

4. Develop a regression model for property price prediction.
 5. Analyze product and regional sales performance.
 6. Examine relationships between sales variables using correlation analysis.
 7. Statistically validate regional sales differences using ANOVA.
 8. Create professional business visualizations.
 9. Translate analytical results into business recommendations.
 10. Develop an implementation plan and measurable KPIs.
-

4. Dataset Description

Three datasets were used in the project.

4.1 Customer Churn Dataset

The customer churn dataset contains **500 records and 9 columns**.

Important variables include:

- CustomerID
- Tenure
- MonthlyCharges
- TotalCharges
- Contract
- PaymentMethod
- PaperlessBilling
- SeniorCitizen
- Churn

The dataset was used to investigate customer retention and churn prediction.

4.2 Property Dataset

The property dataset contains **300 records and 8 columns**.

Important variables include:

- Property_ID
- Area
- Bedrooms
- Bathrooms
- Age
- Location
- Property_Type
- Price

The dataset was used to analyze property price relationships and develop a price prediction model.

4.3 Sales Dataset

The sales dataset contains **100 records and 7 columns**.

Important variables include:

- Date
- Product
- Quantity
- Price
- Customer_ID
- Region
- Total_Sales

The dataset was used to evaluate sales performance across products and regions.

Dataset Summary

Dataset	Records	Main Purpose
Customer Churn	500	Churn analysis and prediction
Property	300	Property price analysis and prediction
Sales	100	Product and regional sales analysis

The customer churn dataset satisfies the project's minimum 500-record dataset requirement. The property and sales datasets are supplementary modules.

5. Data Preparation

Data preparation was performed before conducting the main analysis.

The datasets were loaded using Python and pandas.

The sales Date column was converted to datetime format to ensure that it could be used correctly for time-based analysis.

The datasets were checked for:

- Dataset dimensions
- Column names
- Data types
- Missing values
- Numerical and categorical variables
- Duplicate or inconsistent records where applicable

Cleaned datasets were also stored in the project's data/ folder.

The final project structure separates raw and cleaned datasets to improve reproducibility and organization.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand distributions, relationships, and business patterns.

The project includes multiple visualizations, including:

Customer Churn Visualizations

- Customer churn distribution
- Churn by contract
- Tenure versus churn
- Monthly charges versus churn
- Customer correlation heatmap

Property Visualizations

- Area versus price
- Property price by location
- Price by property type
- Property correlation heatmap

Sales Visualizations

- Sales by product
- Sales by region
- Quantity by product
- Average transaction value by region
- Product-region sales heatmap
- Sales correlation heatmap

These visualizations were used to support the numerical and statistical analysis.

7. Customer Churn Analysis

7.1 Analytical Approach

The customer churn analysis examined:

- Tenure
- Monthly charges
- Total charges
- Contract
- Payment method
- Paperless billing
- Senior citizen status

A logistic regression model was developed to classify customers into churn and non-churn categories.

The data was divided into:

- 80% training data
- 20% testing data

A stratified split was used to maintain the distribution of the churn classes.

Categorical variables were encoded using one-hot encoding, while numerical variables were passed directly to the model.

8. Customer Churn Model Results

The logistic regression model produced the following results:

Metric	Result
Accuracy	93.0%
Precision	75.0%
Recall	54.5%
F1 Score	63.2%
ROC-AUC	0.982

The confusion matrix was:

	Predicted No Churn	Predicted Churn
Actual No Churn	87	2
Actual Churn	5	6

The model correctly classified 93% of the test observations.

It identified **6 of the 11 actual churners**, resulting in a churn recall of **54.5%**.

The model missed **5 actual churners**, represented by the false-negative cases.

The ROC-AUC of **0.982** indicates strong discrimination between churn and non-churn observations in this test sample. However, ROC-AUC should not be interpreted as 98.2% accuracy.

Business Interpretation

The model can provide a foundation for an early-warning churn system. Customers with higher predicted churn probability can be reviewed for possible retention activities.

However, because the model missed some actual churners, threshold tuning or additional modeling could be considered if the business priority is to identify a larger proportion of potential churners.

9. Property Pricing Analysis

9.1 Analytical Approach

The property analysis examined the relationship between property characteristics and price.

The model used:

- Area
- Bedrooms
- Bathrooms
- Age
- Location
- Property Type

A linear regression model was developed.

The dataset was divided into:

- 80% training data
- 20% testing data

Categorical variables were one-hot encoded before training the model.

10. Property Price Model Results

The model achieved:

Metric Result

MAE 2,188,736.34

RMSE 2,907,633.21

R² 0.941

The R² value of **0.941** indicates that approximately **94.1% of the variation in property prices in the test set is explained by the variables included in the model.**

The MAE of **2,188,736.34** represents the average absolute difference between predicted and actual prices in the dataset's price units.

The RMSE of **2,907,633.21** is higher than the MAE because RMSE gives greater weight to larger prediction errors.

Business Interpretation

The model indicates that the selected property characteristics provide substantial predictive information for estimating property prices in this dataset.

The model can therefore be used as a supporting analytical tool for property evaluation.

However, predictions should be reviewed against actual market conditions and should not be treated as the only input for pricing decisions.

11. Sales Performance Analysis

The sales analysis examined:

- Total sales
- Quantity sold
- Product performance
- Regional performance
- Average transaction value
- Product-region performance
- Relationships between quantity, price, and total sales

11.1 Overall Sales Performance

The analysis produced the following results:

Metric	Value
Total Sales	12,365,048

Metric	Value
Total Quantity	478
Average Transaction Value	123,650.48
Average Product Price	25,808.51

These measures provide an overall view of the sales activity recorded in the dataset.

12. Product Performance

Total sales by product were:

Product	Total Sales	Quantity
Laptop	3,889,210	136
Tablet	2,884,340	127
Phone	2,859,394	101
Headphones	1,384,033	48
Monitor	1,348,071	66

Laptop recorded the highest total sales in the dataset, followed by Tablet and Phone.

Laptop also recorded the highest quantity sold.

These findings are descriptive and represent the observed sales records rather than a prediction of future performance.

13. Regional Performance

Regional sales results were:

Region	Total Sales	Average Transaction
North	3,983,635	142,272.68
South	3,737,852	138,438.96
East	2,519,639	132,612.58
West	2,123,922	81,689.31

North recorded the highest total sales and highest average transaction value in the dataset.

West recorded the lowest total sales and average transaction value.

These differences provide useful descriptive information for further investigation, but regional differences should not automatically be interpreted as causal effects.

14. Sales Correlation Analysis

Pearson correlation was used to examine relationships between Quantity, Price, and Total_Sales.

Variable Pair	Correlation
---------------	-------------

Quantity – Total Sales	0.688
------------------------	-------

Price – Total Sales	0.646
---------------------	-------

Quantity – Price	0.008
------------------	-------

The correlation between Quantity and Total_Sales is **0.688**, indicating a positive linear association.

The correlation between Price and Total_Sales is **0.646**, also indicating a positive linear association.

The correlation between Quantity and Price is approximately **0.008**, indicating almost no linear association in this dataset.

Correlation indicates association and does not establish causation.

15. Statistical Analysis – One-Way ANOVA

A one-way ANOVA was conducted to determine whether average sales differed significantly among:

- East
- North
- South
- West

Hypotheses

H₀: The mean sales are equal across all four regions.

H₁: At least one region has a different mean sales value.

Results

Statistic	Result
-----------	--------

F-statistic	2.164
-------------	-------

P-value	0.097
---------	-------

Significance Level 0.05

Decision	Fail to reject H ₀
----------	-------------------------------

Since the p-value of **0.097** is greater than **0.05**, there is insufficient statistical evidence to conclude that average sales differ significantly across the four regions at the 5% significance level.

This does not mean that the regional means are exactly equal. It means that the observed differences are not statistically significant under this test and significance level.

Business Interpretation

Regional decisions should therefore consider additional factors such as:

- Product mix
- Customer behavior
- Pricing
- Marketing activities
- Transaction volume

- Seasonal effects

rather than relying only on observed regional averages.

16. Key Business Insights

The major insights from the project are:

Customer Churn

The churn model achieved strong overall classification performance, but its churn recall of 54.5% indicates that some actual churners were not identified.

Property Pricing

The property regression model achieved an R^2 of 0.941, indicating substantial predictive information from the selected property characteristics.

Product Sales

Laptop generated the highest total sales and quantity in the observed sales dataset.

Regional Sales

North recorded the highest total sales and average transaction value.

Sales Relationships

Quantity and price both showed positive relationships with total sales.

Statistical Validation

The ANOVA did not provide sufficient evidence of statistically significant differences in mean sales across the four regions at the 5% significance level.

17. Business Recommendations

17.1 Customer Retention

The organization can develop an early-warning system using predicted churn probabilities.

Recommended actions include:

- Identify customers with higher predicted churn probability.
- Conduct targeted retention campaigns.
- Review contract and billing experiences.
- Test loyalty incentives or targeted offers.
- Monitor model recall and precision.
- Adjust the classification threshold if the business requires higher churn detection.

17.2 Property Pricing

Recommended actions include:

- Use model-generated price estimates as a supporting reference.
- Consider area, bedrooms, bathrooms, age, location, and property type.
- Compare predicted prices with actual market prices.
- Investigate properties with unusually large prediction differences.

- Retrain the model when sufficient new property data becomes available.

17.3 Sales Optimization

Recommended actions include:

- Monitor product-level sales performance.
- Track quantity and revenue trends.
- Review product-region combinations.
- Use sales patterns to support inventory planning.
- Investigate lower-volume products.
- Monitor average transaction value.
- Evaluate promotional activities using measurable sales KPIs.

17.4 Regional Decision-Making

Since the ANOVA did not find statistically significant differences in average sales at the 5% level, regional strategy should incorporate product mix, customer characteristics, marketing activity, and transaction patterns.

18. Implementation Plan

Phase 1 – Customer Retention

Timeline: Month 1

Actions

- Calculate churn probability for active customers.
- Create risk segments.
- Launch targeted retention campaigns.
- Monitor campaign response.

Success Metrics

- Monthly churn rate
 - Retention rate
 - Churn prediction recall
 - Campaign response rate
-

Phase 2 – Property Pricing

Timeline: Month 1–2

Actions

- Integrate model estimates into property evaluation.
- Generate estimates for new properties.
- Compare predictions with actual market prices.
- Monitor prediction errors.
- Retrain the model periodically.

Success Metrics

- MAE
 - RMSE
 - Prediction consistency
 - Pricing error
-

Phase 3 – Sales Optimization

Timeline: Month 2–3

Actions

- Monitor product sales.
- Review regional performance.
- Analyze product-region combinations.
- Adjust inventory based on demand patterns.
- Evaluate promotional activities.

Success Metrics

- Total sales
 - Average transaction value
 - Product sales
 - Quantity sold
 - Inventory utilization
-

Phase 4 – Continuous Monitoring

Timeline: Ongoing

Actions

- Maintain monthly business dashboards.
 - Track customer churn.
 - Monitor property prediction accuracy.
 - Monitor sales KPIs.
 - Compare predictions with actual outcomes.
 - Retrain models when data patterns change.
-

19. Key Performance Indicators

The following KPIs can be monitored after implementation:

Customer KPIs

- Customer churn rate

- Customer retention rate
- Churn prediction recall
- Churn prediction precision

Property KPIs

- Mean Absolute Error
- Root Mean Squared Error
- R^2
- Difference between predicted and actual price

Sales KPIs

- Total sales
 - Quantity sold
 - Average transaction value
 - Product-level sales
 - Regional sales
-

20. Project Limitations

The analysis has several limitations that should be considered when interpreting the results.

1. The sales dataset contains only 100 records, so sales findings should be validated using a larger dataset.
 2. The property dataset contains 300 records and may not represent all property markets.
 3. The customer churn dataset contains exactly 500 records, so model performance may change with additional data.
 4. Machine learning results are based on the provided train-test split.
 5. Correlation does not establish causation.
 6. Regional descriptive differences should not be interpreted as statistically significant based only on the ANOVA results.
 7. External business factors such as marketing campaigns, competitor activity, economic conditions, and customer satisfaction were not included in the current datasets.
-

21. Conclusion

This project demonstrates an end-to-end business analytics process beginning with data preparation and exploratory analysis and continuing through statistical testing, machine learning, business interpretation, and implementation planning.

The customer churn analysis provides a foundation for identifying customers who may require retention attention. The property pricing model demonstrates the usefulness of property characteristics for estimating prices. The sales analysis provides insights into product performance, regional performance, transaction values, and relationships among sales variables.

The statistical analysis also demonstrates the importance of validating descriptive observations. Although regional sales values differed in the observed data, the ANOVA did not provide sufficient evidence to conclude that average sales differed significantly across the four regions at the 5% significance level.

Overall, the project demonstrates how business data can be converted into measurable insights, analytical models, and practical implementation plans. Future work should focus on collecting larger datasets, incorporating additional business variables, monitoring model performance over time, and validating the recommendations using real-world business outcomes.

22. Technical Environment

The project was developed using Python and Jupyter Notebook.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Scikit-learn
- Jupyter Notebook

Analytical Techniques

- Data Cleaning
 - Descriptive Statistics
 - Exploratory Data Analysis
 - Data Visualization
 - Pearson Correlation
 - One-Way ANOVA
 - Logistic Regression
 - Linear Regression
 - Model Evaluation
-

23. Project Structure

The final project repository is organized as follows:

Complete-Business-Analysis/

```
|  
|— README.md  
|— capstone_analysis.ipynb  
|— requirements.txt  
|  
|— data/  
|   |— customer_churn.csv  
|   |— house_prices.csv  
|   |— sales_data.csv  
|   |— cleaned_customer_churn.csv  
|   |— cleaned_property_data.csv
```

- | └─ cleaned_sales_data.csv
- |
- | └─ reports/
 - | └─ executive_summary.pdf
 - | └─ business_analysis_report.pdf
- |
- | └─ presentation/
 - | └─ business_analysis_presentation.pptx
- |
- | └─ visualizations/
 - | └─ customer_churn_distribution.png
 - | └─ churn_by_contract.png
 - | └─ tenure_vs_churn.png
 - | └─ monthly_charges_vs_churn.png
 - | └─ customer_correlation_heatmap.png
 - | └─ area_vs_price.png
 - | └─ price_by_property_type.png
 - | └─ property_price_by_location.png
 - | └─ property_correlation_heatmap.png
 - | └─ sales_by_product.png
 - | └─ sales_by_region.png

End of Technical Report